25900251 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEBTG200, MG3DSEBBT201 - DEVELOPMENTAL BIOLOGY AND ASSISTED REPRODUCTION TECHNOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

One word or sentence

Answer any **10** questions

Each question carries **2** marks

1. Differentiate between vitelline membrane and zona pellucida in two points. [U] / [CO1]
2. Identify the life stage in women marks the end of reproductive ability. [K] / [CO2]
3. State how many layers are formed by delamination [U] / [CO3]
4. Name the prenatal procedure involves taking a small amount of amniotic fluid for genetic testing [K] / [CO4]
5. Show the the main purpose of fetoscopy [K] / [CO4]
6. Identify the role of LH in the menstrual cycle [K] / [CO1]
7. Name two organisms where parthenogenesis is a regular mode of reproduction. [K] / [CO2]
8. Name the type of blastula lacks a cavity [U] / [CO3]

9. Write one of significance of pedigree analysis? [U] / [CO4]

10. Name the 32 celled stage in cleavage? [U] / [CO3]

11. Name the cells in the seminiferous tubules provide nourishment and structural support to developing sperm? [K] / [CO1]

12. Identify the first milk secreted after childbirth. [K] / [CO2]

[2x10 = 20]

Part B

Short Essay

Answer any **4** questions

Each question carries **5** marks

13. Describe the structural and functional significance of the notochord formed from mesoderm. [K] / [CO3]

14. Examine the importance of ultrasound monitoring during pregnancy. [U] / [CO3]

15. Explain on placental barrier. [U] / [CO2]

16. Write in detail about spermatids [U] / [CO1]

17. Compare fast block and slow block in polyspermy prevention [U] / [CO2]

18. Summarise the applications of fate maps. [U] / [CO3]

[5x4 = 20]

Part C

Essay type questions

Answer any **2** questions

Each question carries **15** marks

19. Discuss the advantages and limitations of IVF ? [U] / [CO4]

20. Explain the process of gastrulation and its significance in embryonic development? [U] / [CO3]

21. Discuss the role of estrogen and progesterone in lactation and mammary gland development [K] / [CO1]

22. Explain the significance of fertilization and the major events involved. [U] / [CO2]

[15x2 = 30]